THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE · MEDIUM LEVEL CHAPTER TEST SET 2

Chapter 11: **Keeping Time with the Skies** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The Hindu lunisolar calendar adds an extra month called: 1

(a) Pausha **(b)** Shravana **(c)** Adhik maas **(d)** Magha

2. The first-quarter Moon is visible in the evening sky and rises at about: 1

(a) sunrise **(b)** sunset **(c)** noon **(d)** midnight

3. A year of 365.25 days in 400 years has: 1

(a) 146097 days (with 97 leap years) **(b)** 146000 days

(c) 146100 days **(d)** 146200 days

4. The Pole Star's altitude above the horizon equals the observer's: 1

(a) longitude **(b)** age **(c)** latitude **(d)** date

5. The Moon is in its waning gibbous phase just after: 1

(a) first quarter **(b)** new moon **(c)** full moon **(d)** no phase

6. Which of the following years is a leap year? 1

(a) 2024 **(b)** 2023 **(c)** 2026 **(d)** 2025

7. A person at the North Pole sees the Pole Star: 1

(a) overhead **(b)** not at all **(c)** in the south **(d)** at the horizon

8. A solar eclipse can occur only on: 1

(a) first quarter **(b)** a new moon day **(c)** any day **(d)** a full moon day

9. Which of the following is NOT a leap year? 1

(a) 2024 **(b)** 2000 **(c)** 1996 **(d)** 1900

10. A person at the equator sees the Pole Star: 1

(a) not at all **(b)** on the northern horizon

(c) in the south **(d)** overhead

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Why do seasons not depend on the distance of the Earth from the Sun? Give one fact. 2

12. How many days are there in 4 solar years including one leap year? 2

13. Explain why the Moon rises about 50 minutes later every day. 2

14. State the position of the Earth in the new-moon and full-moon alignments. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. How are the phases of the Moon different from eclipses? Give three points. 3

16. Explain why the same stars are not visible all through the year. 3

17. Describe how ancient Indian astronomers like Aryabhata contributed to timekeeping. 3

OR

Explain why the Earth's seasons are reversed in the northern and southern hemispheres. 3

18. Explain how the apparent motion of stars can be shown by a long exposure photograph of the night sky. 3

OR

Explain how sunrise and sunset times can be used to find the length of the day and how this changes with seasons. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Explain the origin of seasons with the help of the tilt and revolution, and the day length at the equator and at the poles during the year. 5

OR

Explain the 'day', 'month' and 'year' in terms of the movements of the Earth and the Moon, giving their approximate lengths and the observations used to measure each. 5

20. Compare the solar, lunar and lunisolar calendars with their year lengths and the methods for keeping them in step with the seasons. 5

OR

Explain how the observation of the sky helped farmers in ancient India to decide the times for sowing and harvest. 5

END OF QUESTION PAPER — Total: 40 Marks

MASTERMIND